



Question Set

Question 1

Consider a European option priced from a characteristic function. Define the state variables (log price, characteristic function, damping parameter, quadrature grid, and FFT spacing) and derive the Fourier inversion relation. State the measure, filtration, and tradability assumptions.

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

Question 2

For a European option priced from a characteristic function, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in a European option priced from a characteristic function. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using a lognormal characteristic function evaluated on a grid from -20 to 20, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

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Question 5

For a European option priced from a characteristic function, compute and interpret bump-and-reprice Greeks from the Fourier price. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes Heston or lognormal characteristic-function parameters fit to vanilla prices. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for a European option priced from a characteristic function. Explain how integrability, damping, and payoff transform conditions prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for a European option priced from a characteristic function. Specify numerical inversion checked against Black-Scholes, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements a European option priced from a characteristic function and gets a stable but wrong answer. Explain how aliasing from an inconsistent FFT grid creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for a European option priced from a characteristic function. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of a European option priced from a characteristic function, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of a lognormal characteristic function evaluated on a grid from -20 to 20. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for a European option priced from a characteristic function. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a European option priced from a characteristic function. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for a European option priced from a characteristic function is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for Heston or lognormal characteristic-function parameters fit to vanilla prices are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For a European option priced from a characteristic function, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for numerical inversion checked against Black-Scholes. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that aliasing from an inconsistent FFT grid could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain a European option priced from a characteristic function and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.